Rutherford's Atomic Model Quiz

Physics • Grade 11

Student Name: _____ **Date:** _____ **Score:** ___ / 3

Read each item carefully and answer.

Part A: Choose the right answer

Tick one box.

1. Based on his experiment, Rutherford proposed that atoms are mostly empty space with a dense nucleus. Which of the following best describes the nucleus of an atom?

- ☐ **A.** It is small and contains few particles.
- ☐ **B.** It is small and contains most of the atom's mass.
- ☐ **C.** It is large and contains few particles.
- ☐ **D.** It is large and contains most of the atom's mass.

2. The nucleus contains most of the atom's mass and is composed of protons and neutrons.

- ☐ **A.** Most of the atom's mass is in the nucleus.
- ☐ **B.** Most of the atom's mass is in the electrons.
- ☐ **C.** The nucleus contains only electrons.
- ☐ **D.** The nucleus is mostly empty space.

Part B: True or false

Tick True or False.

3. The scattering of alpha particles in Rutherford's experiment was much greater than expected, indicating the presence of a dense nucleus. This statement is _____.

- [ ] **True**
- [ ] **False**

Answer Key

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1. B. It is small and contains most of the atom's mass. — Rutherford proposed that atoms are mostly empty space with a dense nucleus.

2. A. Most of the atom's mass is in the nucleus. — The nucleus contains most of the atom's mass and is composed of protons and neutrons.

3. True — The scattering of alpha particles in Rutherford's experiment was much greater than expected, indicating the presence of a dense nucleus.